

FILE 095 / DESIGNER

CHINMAY KHAMKAR

Industrial & Automotive Designer

B.Des, MIT-WPU Pune. I build what I draw.

SCAN FOR PORTFOLIO

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01 PROFILE

Final-year design student at MIT-WPU, Pune, working across industrial and automotive design. I enjoy building what I design, from an ultrasonic utensil washer for Indian kitchens and a redesigned air cooler to a traffic signal add-on and an electric conversion of a Royal Enfield Bullet (patent application in progress). I also run a small fabrication workshop, Vahanik Customs.

I'm seeking opportunities in industrial or automotive design where I can take a product from initial user research and conceptual sketches to a functional prototype.

02 EXPERIENCE

Founder and lead fabricator — 2025 – Present
Vahanik Customs, Pune

- Established and manage a home-based workshop specialising in custom motorcycle modifications, chassis repairs and general servicing.
- Fully rebuilt and modified a Honda CBR 250R, covering everything from bodywork, engine and gear upgrades to ECU tuning, suspension modifications and braking system improvements.

Design intern — Summer 2025 · 61 days
Life Transformation Centre, MIT-WPU · Boribel, Daund

- Selected as one of 25 interns to design, fabricate and install 10 theme projects across a 170-acre learning campus, including a 35-foot landmark pylon and wayfinding gateways.
- Served as the primary point of contact between our team and the client's coordinators.
- Welded and assembled road barricades in the workshop, then supervised their on-site anchoring and levelling.

Video editor — Feb – Mar 2025
Carma and Company

- Edited client videos and contributed to their visual branding.

Intern — May – Jul 2021
CRY (Child Rights and You)

- Conducted fundraising and awareness campaigns for child-rights programmes.

03 RESEARCH

eSwitch Dual Drive — 2026 · Ongoing
B.Des research project · Parallel-hybrid CNG–electric retrofit for auto-rickshaws

- The project addresses a key problem faced by auto-rickshaw drivers: significant lost earnings while waiting in CNG filling queues. Conventional electric conversions require removing a functional engine, involve high upfront costs and can limit driving range.
- Developing a bolt-on kit that integrates a low-voltage electric drive alongside the factory-fitted CNG engine, allowing drivers to operate on electric power in city traffic and on CNG for longer distances through a simple dashboard control.
- Combining secondary research and field visits with drivers to plan the installation of the motor, coupling and modular battery within the existing three-wheeler chassis using Fusion 360.

04 EDUCATION

2023 – 2027
B.Des, Product Design
MIT-WPU School of Design, Pune

2023
Class XII, PCB with Psychology
Army Public School, Sagar

2021
Class X
Delhi Public School, Sangrur

05 AWARDS

2024
Indian Army AWWA Award
Southern Command, for the electric go-kart

2022
4th place nationally
All India Science Exhibition, Amity University, for the Eco Jeep

2026
Patent application in progress
Royal Enfield Bullet EV conversion

06 LANGUAGES

English, Hindi, Marathi

07 OUTSIDE THE WORKSHOP

Long motorcycle rides, trekking, robotics, and caring for four large dogs.

09 SKILLS

DESIGN

User research
Concept development
Prototyping
DFM
CMF
System design

SOFTWARE

Fusion 360
Rhino
Photoshop
Illustrator
Premiere Pro
iMovie

WORKSHOP

Arc welding
Metal fabrication
EV conversion
BLDC setup
3D printing

ELECTRONICS

Arduino
IR / ultrasonic / thermal sensors

10 LEADERSHIP

2025 – 26
University Vice President
MIT-WPU

2025 – 26
Coordination Lead
Design XPO 2026

2023 – Present
Class Representative
Product Design, MIT-WPU

2023
School Captain
Army Public School, Sagar

2019
Campus Ambassador
For APS Akhnoor at IIT Delhi

11 CERTIFICATIONS

IIT Delhi · 2019
AI and ML
Automobile Engineering
Introduction to Robotics

IFR · 2021
Basic Robotics
Advanced Robotics

Beep Learning · 2026
Automobile Management
Electric Vehicle Management
Certified Two-Wheeler Service Technician

08 PROJECTS

AUTOMOTIVE AND MOBILITY

Royal Enfield Bullet 350, EV conversion PATENT APPLICATION IN PROGRESS — 2022 – 23

Replaced the original engine with a 48 V high-torque BLDC motor mounted on a custom geared mount. Retained the original gearbox and chain drive, with the battery integrated into the former engine bay. Achieves a range of over 95 km on a 4–5-hour charge.

Electric go-kart AWWA AWARD — 2024

Designed and fabricated a welded steel ladder frame, incorporating a 24 V brushed DC motor with chain drive and repurposed scooter tyres.

Eco Jeep NATIONAL 4TH · AMITY UNIVERSITY — 2019 – 21

Developed a two-seat pedal-electric quadricycle using a steel-tube frame and repurposed bicycle components, featuring Ackermann steering and an electric range of approximately 25 km.

Electric waste-collection tricycle — 2023

Converted a sanitation worker's manual tricycle to a 24 V electric drive system to reduce the physical effort of pushing and pedalling heavy loads.

Mobile Critical Care Unit — 2026 · Team of 4

Developed a concept for a smart stretcher and patient transportation system as part of the Emerging Product Technologies course.

INDUSTRIAL AND SYSTEMS

Kshovik, ultrasonic utensil washer — 2025 – 26 · Team of 5

Designed a tabletop ultrasonic washer for whisks, graters and strainers, inspired by the form of traditional Indian cooking pots. My contributions included research, mock-ups, a functional proof of concept and 3D modelling. The project was subsequently pitched to investors.

Smart Traffic Management System — 2025 · Team of 3

Developed an add-on system for existing traffic signals that uses a thermal camera to count vehicles and dynamically adjust green-light durations, with an override mechanism for ambulances. Following 30 user interviews, we built a functional traffic signal mock-up and a junction model.

Air cooler redesign (Symphony Diet 35T) — 2025 · Solo

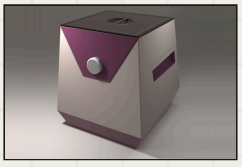
Disassembled the cooler, identified 14 design issues and surveyed 30 owners. Water spillage emerged as the most common complaint (77%). The redesign addresses this through a top-fill mechanism, a visible water-level indicator and lockable wheels.

Library on Wheels — 2025 · Team of 7

Design students rarely made the long trip to MIT-WPU's central library. We designed a service in which a mobile library cart brings the books and journals students need to the School of Design every week. Deliverables included an 87-page research report, a cart prototype and storyboards.

Additional projects include furniture made from scrap tyres, an ultrasonic sink dishwasher, brand studies of Honda BigWing and the Volkswagen Beetle, and a heritage documentation project on Vishrambaug Wada.


01 BULLET EV


02 KSHOVIK


03 ECO JEEP


04 GO-KART


05 AIR COOLER


06 LIBRARY CART


07 TRAFFIC MANAGEMENT


08 LTC INTERNSHIP